

DS 2 Experiment

Code:

```
#include <stdio.h>

void accept(int a[10][10], int m, int n)
{
    printf("Enter values of sparse matrix:\n");
    for(int i = 0; i < m; i++)
    {
        for(int j = 0; j < n; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }
}

int convert(int a[10][10], int m, int n, int s[10][3])
{
    s[0][0] = m;
    s[0][1] = n;

    int k = 1;

    for(int i = 0; i < m; i++)
    {
        for(int j = 0; j < n; j++)
        {
            if(a[i][j] != 0)
            {
                s[k][0] = i;
                s[k][1] = j;
                s[k][2] = a[i][j];
                k++;
            }
        }
    }

    s[0][2] = k - 1;

    printf("\nSparse Matrix:\n");
    for(int i = 0; i < k; i++)
    {
        printf("%d\t%d\t%d\n", s[i][0], s[i][1], s[i][2]);
    }
}
```

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    return k - 1;
}

void transpose(int s[10][3], int m, int n, int t[10][10], int q)
{
    printf("\nSimple Transpose:\n");

    int k = 1;

    t[0][0] = n;
    t[0][1] = m;
    t[0][2] = q;

    for(int i = 0; i < n; i++)
    {
        for(int p = 1; p <= q; p++)
        {
            if(s[p][1] == i)
            {
                t[k][0] = s[p][1];
                t[k][1] = s[p][0];
                t[k][2] = s[p][2];
                k++;
            }
        }
    }

    for(int i = 0; i <= q; i++)
    {
        printf("%d\t%d\t%d\n", t[i][0], t[i][1], t[i][2]);
    }
}

void fasttranspose(int s[10][3], int f[10][10], int m, int n, int q)
{
    printf("\nFast Transpose:\n");

    f[0][0] = n;
    f[0][1] = m;
    f[0][2] = q;

    int rowt[10] = {0};
    int starti[10];

```

```

for(int i = 1; i <= q; i++)
{
    rowt[s[i][1]]++;
}

starti[0] = 1;

for(int i = 1; i < n; i++)
{
    starti[i] = starti[i - 1] + rowt[i - 1];
}

for(int i = 1; i <= q; i++)
{
    int j = starti[s[i][1]];

    f[j][0] = s[i][1];
    f[j][1] = s[i][0];
    f[j][2] = s[i][2];

    starti[s[i][1]]++;
}

for(int i = 0; i <= q; i++)
{
    printf("%d\t%d\t%d\t%d\n", f[i][0], f[i][1], f[i][2]);
}
}

int main()
{
    int m, n, choice;
    int a[10][10], s[10][3], t[10][10], f[10][10];

    printf("Enter rows and columns: ");
    scanf("%d%d", &m, &n);

    accept(a, m, n);

    int total = convert(a, m, n, s);

    do
    {

```

```

printf("\n----- MENU -----");
printf("\n1. Simple Transpose");
printf("\n2. Fast Transpose");
printf("\n3. Exit");
printf("\nEnter choice: ");
scanf("%d", &choice);

switch(choice)
{
    case 1:
        transpose(s, m, n, t, total);
        break;

    case 2:
        fasttranspose(s, f, m, n, total);
        break;

    case 3:
        printf("Exiting...\n");
        break;

    default:
        printf("Invalid Choice\n");
}

} while(choice != 3);

return 0;
}

```

Output:

```

Enter rows and columns: 3
3
Enter values of sparse matrix:
0
0
7
7
0
0
0
9
0

```

Sparse Matrix:

3	3	3
0	2	7
1	0	7
2	1	9

----- MENU -----

1. Simple Transpose
2. Fast Transpose
3. Exit

Enter choice: 1

Simple Transpose:

3	3	3
0	1	7
1	2	9
2	0	7

----- MENU -----

1. Simple Transpose
2. Fast Transpose
3. Exit

Enter choice: 2

Fast Transpose:

3	3	3
0	1	7
1	2	9
2	0	7

----- MENU -----

1. Simple Transpose
2. Fast Transpose
3. Exit

Enter choice: 3

Exiting...